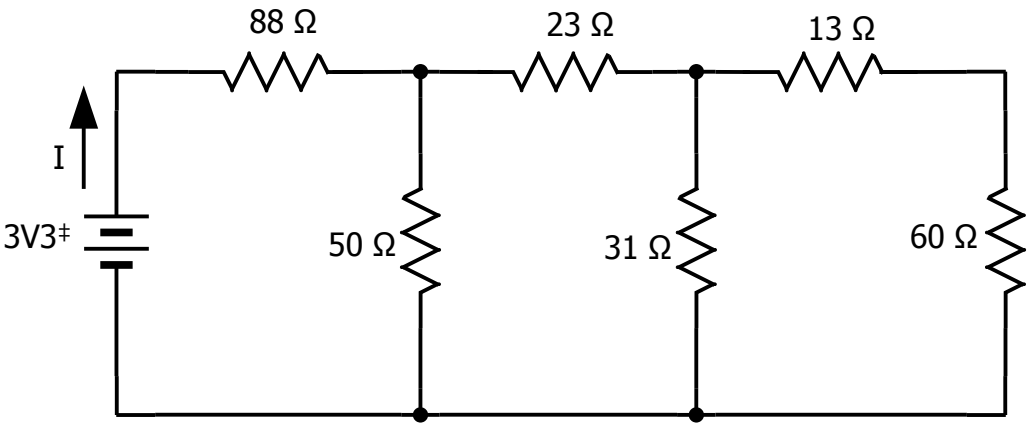
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1. Using your knowledge of series and parallel equivalents that you learned on p. 23 of the Week 1 lab, determine the current I.

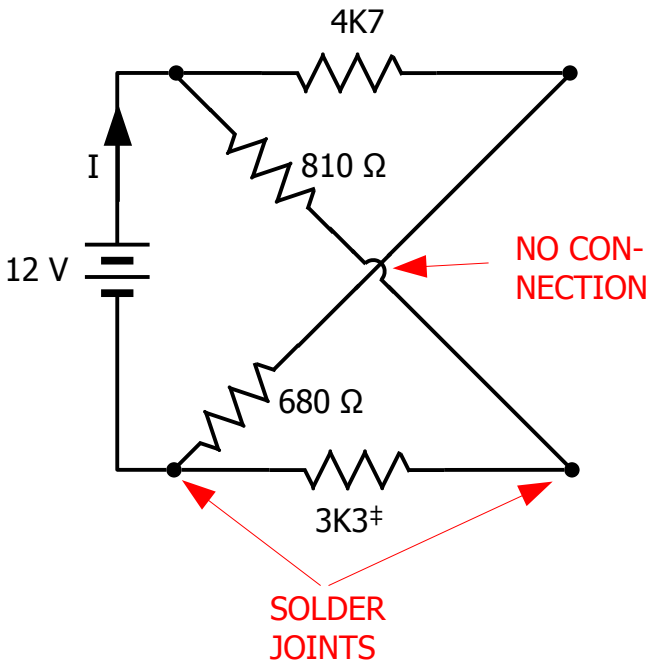


$I = 0.0296$ A

‡ 3V3 is shorthand for 3.3 V. You will see this often on schematics.

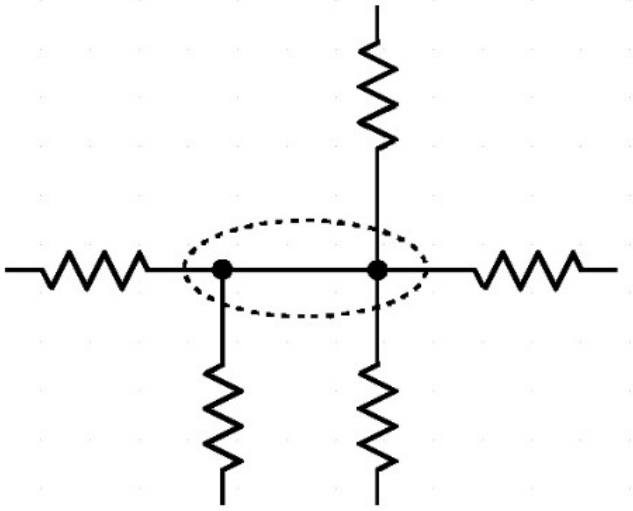
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2. Using your knowledge of series and parallel equivalents, determine the value of I.



$I = 5.15$ mA

3. This question may require a little digging on your part.



The circuit fragment inside the dotted ellipse is:

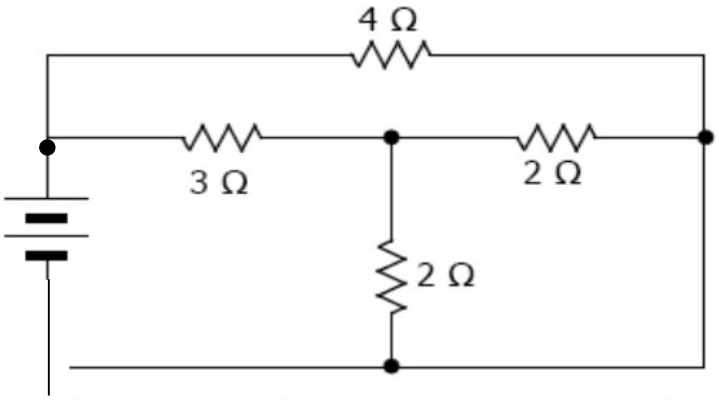
Choose one answer.

- a. not a node
- b. 1 node
- c. 2 nodes

‡ 3K3 is shorthand for 3.3KΩ. You will see this often on schematics.

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4.



This circuit can be reduced to a voltage source and one resistor. What is the value of that one final resistor?

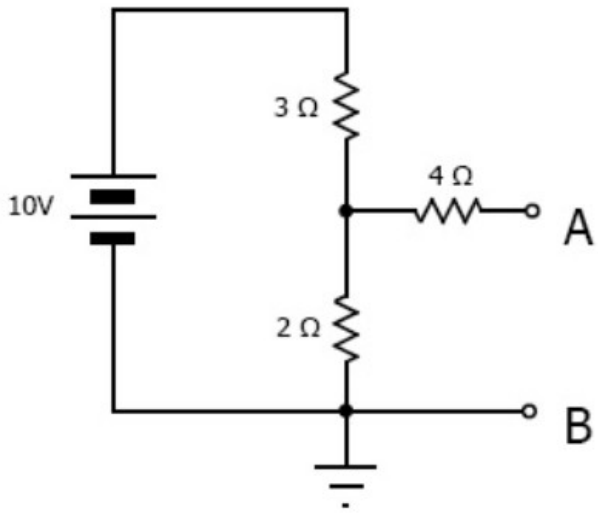
Choose one a. 3.0 Ω

Challenge: try to do this one in your head: no paper/pencil, no calculator/computer. If you understand series and parallel, you can do it.

$R_{tot} = 2\,\Omega$

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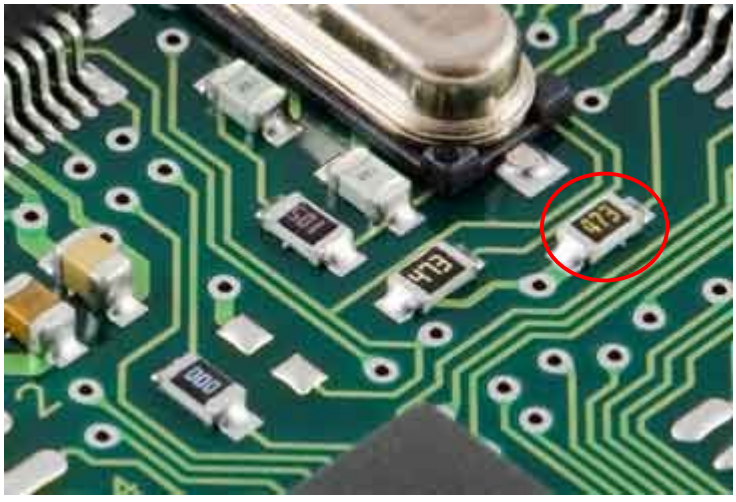
5. What is the voltage across the 4 Ω resistor? NOTE: you may consider this to be a trick question, but it has real meaning.



Choose one answer.

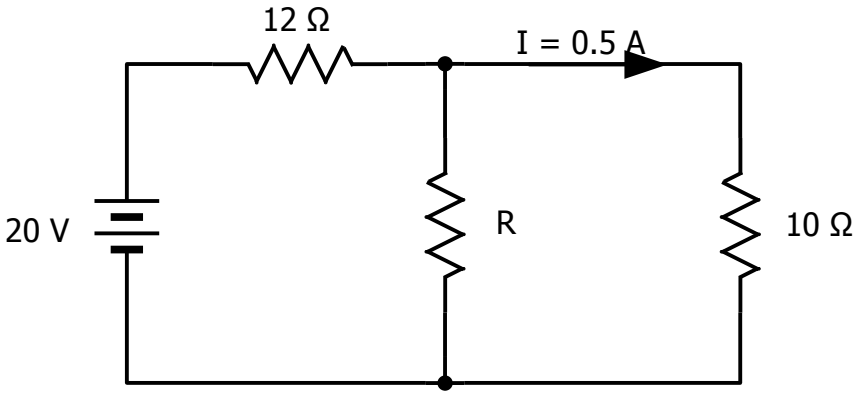
- a. 2.67 V
- b. 0.00 V
- c. 5.71 V
- d. 10.00 V
- e. 4.00 V

6. This is the picture of a surface mount resistor (labeled 473) on a PC board. Assuming that the three numbers correspond to the first three colors of a regular resistor, what is the resistance of this surface mount resistor?



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7. Using your knowledge of series and parallel equivalents plus the Voltage Divider equation, determine the value of R.



$R = 6.67\ \Omega$

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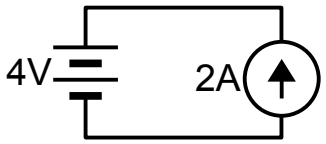
8.

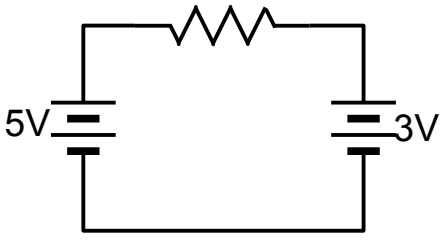
a. Is the 4V battery providing or absorbing power?

b. Is the 2 A current source providing or absorbing power?

c. Is the 5V battery providing or absorbing power?

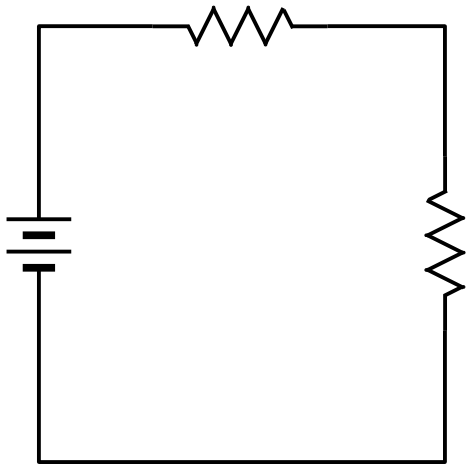
d. Is the 3V battery providing or absorbing power?





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9. Indicate the direction of conventional current.



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10. Draw the I-V curve for a resistor of value R. Indicate the slope of the curve.

